Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

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German Aerospace Center (DLR), Institute of Vehicle Concepts, Oberpfaffenhofen
Attn: Dr. Jonathan Brembeck

Application: Vacancy-ID 6202, model- and learning-based control for by-wire vehicles

Dear Dr. Brembeck,

I am applying for the position on model- and learning-based simulation and control methods for robotic electric vehicles (Vacancy-ID 6202). I am a mechatronics and robotics engineer doing my PhD at Aalborg University, with my thesis expected to be finished in March. What I enjoy most is the step where a control method leaves the computer and has to work on real hardware on a test bed. That is exactly what this role is about.

Control of nonlinear mechatronic systems has been part of my work from the start. In my Master's at CSIR-CMERI I worked on a pneumatic artificial muscle, which is a strongly nonlinear actuator. I designed the outer PID loop of a neural network-based gain-scheduled controller in MATLAB/Simulink and implemented it in LabVIEW to drive the muscle, reaching 0.3 to 0.78% position error (IEEE CONECCT 2022). In my PhD I designed a hybrid shoulder exoskeleton, optimised the design with a genetic algorithm, and tested it on a physical setup and with 12 participants. In a team of 3 I helped with the equations, such as the Fourier fit, for an Assist-as-Needed controller. I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms, estimated its stiffness with less than 10% error and quantified human arm impedance.

I am used to testing methods on real hardware and judging them with numbers. I integrate motors, encoders and force sensors, build test setups, and implemented an Arduino-based trigger system to synchronise robot motion with Qualisys motion capture. My work on the exoskeleton won the Best Paper Award at IFToMM ISRM 2026, and I am comfortable writing for journals and conferences.

I want to be honest about the gaps. I have not yet developed reinforcement learning or LLM-based toolchains myself, and I have not worked on vehicles. My Python is basic, while MATLAB/Simulink is my main tool. I learn quickly when I can test ideas on a real system, and I would really like to build these skills in your team.

Thank you for considering my application. You can see my work at arunabha22.github.io/arunabha-majumder.

Kind regards,
Arunabha Majumder